

Minor Resistance: The Everyday Politics and Power Dynamics of Assistive Technology Adoption

Stacy Hsueh

Paul G. Allen School of Computer Science and Engineering
University of Washington
Seattle, Washington, USA
shsueh@cs.washington.edu

Danielle Van Dusen

Occupational Therapist & Assistive Technology Specialist
Seattle, Washington, USA
daniellevandusen@gmail.com

Anat Caspi

Paul G. Allen School of Computer Science and Engineering
University of Washington
Seattle, Washington, USA
caspi@cs.washington.edu

Jennifer Mankoff

Paul G. Allen School of Computer Science and Engineering
University of Washington
Seattle, Washington, USA
jmankoff@cs.washington.edu

Abstract

In accessibility research, the choice to adopt or abandon assistive technologies (AT) is often taken as a proxy for functional fit: to adopt is to confirm a good fit between device features and individual needs, whereas to abandon is to signal poor fit. While useful for orienting design, we argue that this framework is ill-equipped to account for the sociopolitical forces that shape AT use in historically underserved communities. In this paper, we propose a power-aware framework that recasts adoption not as transparent expression of fit, but as situated negotiation of power. Drawing from an eight-month ethnographic study at a Seattle-based nonprofit, we examine how low-income, racially diverse, and disabled families navigate institutional practices that impose normative expectations around disability and AT use. We introduce the concept of minor resistance to describe the everyday ways people exercise agency in response to power dynamics that make access costly. We argue that this shift in analytical lens reframes the goal of accessibility from optimizing use to lowering the cost of choice. We conclude with implications for how designers can support community-engaged responses to structural barriers by centering self-determination.

CCS Concepts

• **Human-centered computing** → **Accessibility theory, concepts and paradigms**; **Empirical studies in accessibility**.

Keywords

Assistive Technology, Power, Social Inequity, Minor Resistance, Empowerment, Intersectionality, Accessibility

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1 Introduction

For more than two decades, accessibility researchers have been interested in facilitating successful adoption of assistive technology (AT) used by people with disabilities [59, 90, 97]. In popular discourse, AT is figured as a technology of empowerment: “*For most of us, technology makes things easier. For a person with a disability, it makes things possible*”, says disability rights activist Judy Heumann. From speech-generating devices to tactile displays, ATs have facilitated a wide range of activities and is used extensively by people with disabilities. It is thus no surprise that a significant body of accessibility literature is dedicated to understanding barriers to AT adoption and reasons for AT abandonment [59, 77, 90, 97], as well as developing ways to address these challenges, such as designing ATs that are more customizable [24] or easier to use [28]. Upon closer examination, the most enduring narratives across these works usually involve improving the *fit* between personal attributes (e.g. needs, goals, values) and technological attributes (e.g. functionalities, aesthetics). Within this framework, adoption is treated as a proxy for fit: adopting indicates good fit, and abandoning reveals poor fit.

This emphasis on fit frames empowerment as a technical project — a matter of getting the right fit and the fit “right”. The assumption is that goodness of fit is a reliable predictor of empowerment. However, such framing obscures the sociopolitical factors that influence how and whether technology can truly empower. As researchers have shown, promises of empowerment through AT vary widely depending on one’s socioeconomic circumstances and often fall short for working-class and low-income communities [5]. In fact, people will use an AT even when it is unusable, because they are structurally compelled to switch to a new device [73, 89]. There is also growing evidence that the process of seeking accommodations can itself be burdensome or harmful [81, 99], suggesting that non-use is a legitimate and strategic response to structural violence. These burdens are often more pronounced for historically underserved¹ communities, who face systemic inequalities that

¹Throughout this paper, we use *underserved* to refer to communities in U.S. contexts with uneven access to resources such as healthcare, education, public services, and

amplify the cost of access [81] through stigmatization, institutional gatekeeping, or prior experiences of harm.

These critiques and analyses suggest a need to go beyond the current *fit* model of AT adoption, as it is ill-equipped to capture the day-to-day realities that underserved communities grapple with. It assumes that use is always desirable, that non-use is a problem to fix, and that the choice is freely made. Instead, taking cues from Liboiron’s notion of “compromised agency” [62], we argue that the *choice* to use — or not use — an AT is not a straightforward response to fit but a strategic negotiation with external forces that compel or foreclose use. The choice model challenges the notion that AT adoption is an apolitical process of making technology work for users, showing instead that uneven distribution of resources and power asymmetries texture adoption in ways that cannot be reduced to a simple correspondence between individual preferences and device features. Empowerment, then, is not just the ability to effectively use a device, but the ability to exercise choices within the specific constraints and opportunities of one’s social location. This shifts design from problem-solving to analyzing, critiquing, and intervening in the systems and relations that shape what choices are possible.

To develop a framework that tackles accessibility as a question of power in addition to design, we conducted an eight-month ethnography at a local Seattle-based non-profit providing AT and disability services support to low-income disabled families. Our research questions are two-fold: (1) How do power relations shape AT access and use? (2) How do families’ responses to these power relations influence their patterns of AT use? In this paper, we approach ATs not as devices existing in a vacuum, but as part of a *sociotechnical assemblage* of social practices, institutions, and technological affordances [26]. To further account for how agency is exercised within systems of domination, we draw on theories of everyday resistance [92] to examine how individuals and communities navigate and push against the structural conditions that shape access and their choices.

Building on this, we introduce the concept of *minor resistance*: everyday strategies people develop to evaluate and mitigate the complex costs of access-related choices, contesting and reshaping the conditions of their participation in asymmetric power relations. By foregrounding minor resistance, we aim to reframe accessibility from a focus on what technologies can enable or augment (i.e. top-down interventions) to how they might support and amplify the existing ways people assert agency within disempowering systems (i.e. bottom-up strategies). This means shifting design away from optimizing use toward lowering the cost of choice.

We begin by situating our work within studies of power, empowerment, and resistance, drawing on the notion of “power-to” and theories of grassroots resistance to develop our concept of *minor resistance*, which anchors our analysis. Informed by ethnographic insights, we then identify ways to interrogate power assumptions in AT adoption and reorient accessibility research. This paper makes the following contributions:

- We provide an analysis of how power operates in everyday AT practices of low-income, disabled families.

- We introduce the concept of *minor resistance* to describe the everyday strategies families use to navigate the frictions created by rigid systems as they integrate ATs into their lives.
- We pose critical questions to guide accessibility researchers in examining power dynamics in AT contexts.
- We offer implications of minor resistance for reorienting design practice around collective forms of access.

2 Theoretical Background

In this section, we outline the theoretical foundations of the paper. Specifically, we draw on Jo Rowlands’ work on power and empowerment [85] to examine power dynamics in accessibility research. We also develop our concept of *minor resistance* from theories of grassroots resistance.

2.1 Putting Power Back in Accessibility

In accessibility research, empowerment is often invoked without a clear articulation of what *power* is. Jo Rowlands, in her examination of empowerment [85], argues that any meaningful discussion of empowerment must begin with a holistic account of power dynamics. It requires grappling with the broader social structures that shape our lives, recognizing power not only as an individual capacity but also as systemic forces that enable or constrain action.

One way to understand power is through its four instantiations, originally proposed by Rowlands [86] and later taken up by numerous scholarship [39, 107]. The first type, “power-over”, refers to the traditional notion of power as domination and control, enacted through hierarchical relations that produce oppression and inequality. By contrast, “power-to” highlights the capacity to act and effect change in response to oppression. “Power-within” relates to an individual’s sense of self-worth, while “power-with” describes the collective power that emerges through collaboration and relationships.

In AT literature, empowerment typically refers to enabling users to create AT for themselves [16, 51, 80], increasing independence [64, 93], and improving performance in daily tasks [36]. This framing of empowerment emphasizes self-sufficiency and self-reliance. Complementing it, researchers have proposed an alternative view of empowerment rooted in interdependence [11, 108], which describes the ways access is created through “relationships between people, ATs, and environments” [11]. These two distinct conceptions of empowerment are also discussed by Parent [76]. Viewing these forms of empowerment through Rowlands’ lens of power, we see that the independence framing can be mapped to *power-within*, which is an individual’s ability to achieve empowerment on their own, whereas the interdependence framing maps to *power-with*, which refers to the shared power that grows out of cooperative practices. The notion of minor resistance introduced in this paper can be understood as a third form of empowerment, *power-to*, which captures the “unique potential of every person to shape [their] life and world” [106].

In this paper, we explore power-to as a frame for designing ATs that support people in self-assertion against systems of power. Examples of people rebelling against hostile systems or structures abound in disability communities. Consider, for example, activists

social support, including low-income, disabled, racialized, and otherwise minoritized populations [1].

in London who poured cement on “anti-homeless” spikes installed by a supermarket to curb “antisocial behaviors” such as smoking [78]. Artist and design researcher Sara Hendren would call this a form of design activism which “uses the language of design to create political debate” [45]. Building on this, we foreground minor resistance strategies observed in this study to articulate a design practice that (1) helps people live with, resist, and subvert inflexible systems, and (2) helps researchers recognize, understand, and support minor resistance as a pathway to transforming power relations in design.

2.2 “Weapons of the Weak”: Conceptualizing Minor Resistance

The concept of grassroots resistance, as explored in sociology and anthropology, provides a lens for understanding actions often interpreted as resistance to AT adoption. A central reference is political scientist James C. Scott’s notion of everyday resistance, which he describes as a “weapon of the weak” [92]. He characterizes it as “an integral part of the small arsenal of relatively powerless groups” [92], exhibiting behaviors such as “foot-dragging”, “feigned ignorance”, or “sabotage”. Scott argues that political scientists have historically neglected the diverse political strategies of marginalized groups, as these actions tend to be more subtle and easily overlooked. He challenges the traditional narrative that portrays vulnerable groups as apathetic or too overwhelmed to engage politically, showing instead how they participate in politics through quotidian ways of evading and manipulating authority [92]. In his view, these everyday forms of resistance provide vital means for marginalized groups to express political voice [92].

Our notion of minor resistance combines Scott’s concept of everyday resistance with recent interdisciplinary work that situates resistance within intersectional power dynamics. Such resistance can take forms ranging from covert actions to organized movements [8, 63]. Following Johansson et al. [54], resistance unfolds “in a process of social interaction” involving three key players, and in the context of our study, they are: (1) *individuals engaging in resistance*, or the disabled families, (2) *agents of power*, such as insurance companies, hospitals, and school districts, and (3) *observers or intermediaries*, like the non-profit offering AT support services. This framing highlights the interplay between institutional power and individual agency. In this paper, power is conceptualized as operating on three levels — “muscle power”, “agenda-setting power”, “discursive power” [65] — and agency is understood as “power-to” [107].

3 Related Work

3.1 Pitfalls of AT Adoption and Abandonment Narratives

Research on AT adoption and abandonment spans multiple disciplines, from HCI [112] to education [7, 68], psychology [17, 91], and occupational therapy [23, 41]. Each foregrounds different explanations for why ATs are taken up or discarded. In HCI and accessibility research, two dominant frameworks have shaped how these dynamics are understood: the *Technology Acceptance Model* (TAM) and the *Matching Person and Technology* (MPT) model. In

TAM [25], technology adoption depends on the degree to which technology meets users intentions, as measured through “perceived usefulness”, “perceived ease of use”, and “behavioral intention”. It has been widely applied to accessibility contexts to identify and validate design factors influencing adoption, such as Djamasbi et al.’s [32] work on web accessibility for blind users and Franz et al.’s [38] study of mobile accessibility for older adults. The MPT model, on the other hand, frames AT adoption as the process of matching user preferences, environmental characteristics, and device usability. It has informed HCI research on the social and cultural dimensions of AT use, such as social stigma surrounding AT use in public spaces [97] and cultural factors affecting attitudes towards AT [82]. Collectively, TAM and MPT provide complementary and overlapping lenses for understanding AT adoption: whereas TAM attends to alignment between technology and user goals, MPT considers the holistic fit of technology with an individual’s lifestyle, needs, and context. Both models ultimately treat AT adoption itself as the desired outcome.

However, looking at AT use through the lens of adoption and abandonment tends to filter out the everyday creative problem-solving people engage in when encountering inaccessible ATs. For example, Kane et al. [55] showed that people with visual and motor disabilities devise myriad strategies to adapt inaccessible devices to their needs. Likewise, Borodin et al. [14] documented a range of browsing strategies that screen reader users employ to navigate inaccessible websites. Jacobson [53] further showed that individuals customize assistive devices not only for practical reasons but also to manage social perceptions of AT use. These studies suggest that the boundary between adoption and abandonment is far more fluid than traditional models imply, as users continually adapt, modify, and work around their assistive devices.

Researchers have begun to challenge adoption and abandonment as taken-for-granted metrics of success or failure. Dawe [27], for example, questions the usefulness of having a single adoption rate for all AT, noting that such a metric ignores “large differences among the devices and the populations that use them”. Not only does adoption rate lack analytical granularity, it also “fails to reveal the ways and contexts in which technology is used” [27]. Taking this critique further, more recent work [98] has called for a rethinking of abandonment in clinical contexts: rather than interpreting non-use as failure, the authors suggest recognizing it as a deliberate choice made by individuals and families “to use a full range of multimodal communication [that better] meet[s] their own needs”. Similarly, Seymour [94] shifts the focus from uncovering reasons for abandonment to examining “the values and beliefs of disabled people as they choose to engage with or to abandon technology”. These studies recast abandonment not as a barrier to overcome, but as a lens for understanding the deeper motivations and everyday realities that structure AT use. However, relatively little is known about how disabled communities in under-resourced settings navigate these choices. Our work examines AT use in contexts of precarity, where people with disabilities face additional structural barriers and resource constraints.

3.2 AT Use in Historically Underserved Communities

HCI and accessibility scholarship has increasingly drawn attention to practices and challenges faced by people living in contexts of economic precarity, resource scarcity, and social exclusion. These settings are marked by interlocking forms of systemic oppression; from infrastructures that sustain racial disparities to institutional practices that reproduce ableism, they present unique challenges for equitable AT experiences and outcomes. Addressing these challenges requires a situated understanding of the social and structural contexts in which these communities live.

For example, researchers studying social media use among low-income blind people in India found that a novel voice-based platform held widespread appeal compared to mainstream apps like WhatsApp, Facebook, or Twitter, suggesting that designs responsive to local cultural and socioeconomic conditions may support more meaningful adoption [105]. Research in the Global South [9, 87] has also shown that intermediated technology use — relying on others to access technology — is a common practice among minoritized populations and should be accounted for in design. Pal et al. [74] examined the differences between public transit systems in Seoul and Bangalore to appeal for context-specific solutions for blind smartphone users. In a study with mobility aid users in low-resource settings, Oldfrey et al. [72] analyzed three distinct models of repair practices that can be leveraged to improve AT provision.

While prior work has cataloged critical barriers to AT use, it has rarely situated them within broader institutional contexts and systems of oppression. Our work extends this scholarship by examining how power valorizes some ATs and not others, governs access, conditions everyday use. Tracing these dynamics, we argue, is necessary for guiding design toward supporting minoritized communities in building power within exclusionary systems.

3.3 AT Through a Sociotechnical Lens

HCI researchers, accessibility scholars, and critical designers have interrogated the politics of ATs, from computer vision systems for visual accessibility [12], to prosthetics [46], cochlear implants [95], and disability dongles [52]. These works demonstrate how technoableism [96] frames disability as undesirable and technology as a means of restoring normative ways of being. ATs can also reinforce sociolinguistic hierarchies: for example, most augmentative and alternative communication (AAC) devices are developed for English and provide limited support for people with complex communication needs [6].

While computational artifacts can embody and perpetuate particular worldviews [21, 111], ATs extend beyond devices; they also include the “services necessary to get and use the devices, including assessment, customization, repair, and training” [2]. HCI research has largely focused on AT as individual devices, yet a service-oriented perspective — such as examining ecosystems of care around 3D printed AT [88] — makes visible how these technologies are embedded in larger sociotechnical systems. With this framing, biased notions about disability and bodily norms are inscribed not only in AT design features but also reproduced through the institutional processes that regulate access and use. For example, speech language pathology literature has shown that AT provision’s close

coupling with medical assessments can sometimes result in over-identifying non-English speakers as language disordered due to the lack of testing resources in their native languages [15, 22, 75]. Beyond diagnostic practices, communications scholar Meryl Alper argues in *Giving Voice* [5] that advantages of AAC devices have accrued disproportionately to those who are socioeconomically privileged. She calls for studying AT as a sociotechnical practice, contesting sweeping claims of empowerment and revealing how AT is entangled with culture, law, and policy in ways that reinforce structural inequalities.

Existing scholarship has demonstrated how biases and inequalities are built into AT design and provision, but how marginalized families navigate, adapt, and push back against these systems remain understudied. In our work, we examine how families develop everyday strategies that allow them to persist through conditions of deep instability, while addressing immediate needs in ways that sustain their autonomy.

4 Methodology

To understand how disabled families in under-resourced communities in the U.S. navigate and negotiate AT services, we conducted an ethnographic field study at a Seattle-based non-profit that provides AT support services to low-income and racially diverse communities.

4.1 Data Collection

We conducted eight months of ethnographic fieldwork (February 2024 to September 2024) at Open Doors for Multicultural Families that included interviews and observations with staff and clients. We also collected archival materials to situate these experiences within the broader landscape of AT practices and infrastructures.

4.1.1 Field site. Open Doors for Multicultural Families (ODMF) is a non-profit organization that helps families with disabilities navigate disability services and obtain access to ATs. The families served are primarily immigrants, refugees, and people of color from diverse cultural backgrounds, including African American, African, Asian, Southeast Asian, Hispanic/Latinx, Middle Eastern, and Eastern European communities. 89-90% of the families served come from low- to very low-income households and may have limited English proficiency. The organization provides programs for all age groups, from infants and children to adults and seniors. Its AT program offers consultation, training, and resources, along with a lending library that features a wide range of low- to high-tech assistive devices.

4.1.2 Observations and ethnographic field notes. The first author worked at ODMF for seven months as an intern, where she documented first-hand accounts of case management, AT consultations, organizational workflows, and lending library activities. During this time, she spent 5-10 hours a week observing and participating in workshops (e.g. making DIY AT, navigating AT services, and Open House), consultation sessions, a local high school visit, the Friendship Circle program, information sessions, and staff meetings. She also helped with lending library inventories and researched new assistive devices to add to the library.

As part of this fieldwork, 12 AT consultation sessions were observed. In these sessions, the AT specialist (second author), sometimes alongside case managers, interviewed families to assess needs and recommend devices for short-term loan from the lending library. Detailed field notes were taken both during and immediately following each session to capture interactions.

4.1.3 Interviews with case managers and families. We conducted 22 interviews at ODMF with staff and clients: five with case managers, three with program managers, 13 with families enrolled in the AT program, and one with a speech language pathologist (SLP). Interviews lasted approximately 90 minutes on average, ranging from 60 to 110 minutes. These interviews were complemented by six months of extensive discussions with an AT specialist and another intern, during which the first author took detailed field notes, as well as additional informal conversations with ODMF staff.

The interviews with families included at least one parent (with or without disability) and a child with disability. In most cases (12 families), the parent sought services for their child, whose ages ranged from 3 to 22 years, with 3 being the most common age. The disabilities represented included cognitive disabilities (10 participants, predominantly non-verbal autism), mobility impairments (2 participants), and hearing impairment (1 participant). In all interviews, we spoke with the parent, though in two cases the child was also present. When children participated, they responded to simple yes/no questions using a communication device. For six participants whose primary language is Spanish, we used LanguageLine for phone interpretation, which is the main interpretation service used at ODMF.

For participant coding, case managers were labeled CMx, program managers PMx, families Px, and the speech language pathologist SLP.

4.1.4 Documents and webinars on cultural AT and DIY AT. We collected archival materials shared by ODMF staff, focusing on resources related to AT services, including policy and funding documents, administrative forms, informational guides, webinars, and podcasts. These materials provided insight into both industry understanding of AT and on-the-ground realities of its use, drawing a holistic picture of how devices and services move through government agencies and intersect with regulations. While we did not formally analyze these materials, they served as important contextual resources that informed our understanding of dominant discourses around AT access, policy, and practice.

4.2 Data Analysis

Following an inductive, grounded theory approach [18], we analyzed our data thematically to identify recurring patterns of power and resistance across participant narratives. The first author kept detailed reflexive memos from field notes and interviews, capturing emergent insights alongside continuous engagement with feminist, intersectional, and postcolonial theories. These memos were reviewed together during weekly meetings, where we collaboratively generated initial codes and observations. Emergent themes included institutional logics embedded in AT services and tensions between bureaucratic and lived timelines. Through iterative rounds of analysis, we refined our interpretations, culminating in our focus

on minor forms of resistance and generative frictions that arise at the intersection of institutional expectations and everyday life.

To develop a power-aware lens for interrogating AT use, we analyzed (1) the sites and mechanisms through which power is exercised across the AT provision pipeline, and (2) the strategies families use to assert their selfhood and values. We refer to these strategies as *minor resistance* – a form of counter-politics enacted not through overt protest but through tactical, everyday interventions. These practices reflect Sara Ahmed’s notion of “willfulness” [3]: acts of disobedience in which people persist through systems not built for them.

4.3 Positionality

The data collection and analysis were influenced by our experiences as accessibility researchers and practitioners. The second author brings professional field experience as an AT specialist. Three of the authors have experience caring for family members and interacting with many of the same systems encountered by the families in this study. As a team, we hold varied relationships to race, class, and disability. While none of us shares the full set of social positions held by the families, all of the authors are first- and second-generation immigrants, three are multilingual, and two are parents of disabled children. Although we are in positions of relative privilege as researchers and practitioners within U.S. institutions, we also contend with structural barriers similar to those surfaced in our fieldwork. Throughout this work, we remain attentive to our solidarities and differences with the families we engaged with. We also place strong emphasis on grounding our empirical findings in relevant theoretical frameworks [44], drawing particularly on studies of everyday resistance to understand how agency emerges in response to top-down processes.

5 Results

In this section, we present the concept of minor resistance and contextualizing it within a broader terrain of power dynamics. We then identify four types of minor resistance.

While our findings focus on schools, hospitals, and insurance companies as the proximate sites through which families navigate AT access, these institutions are shaped by the larger U.S. sociopolitical landscape marked by racialized rhetoric, austerity politics, and escalating Big Tech and state power. Though a full accounting of these forces lies beyond the scope of our current study, we see this context as critical for situating the more immediate power relations described in the following section.

5.1 Framing Minor Resistance: Power Situations in Everyday Disabled Lifeworlds

Before discussing specific practices of minor resistance, we first examine the cross-cutting structural forces that influence AT use, illustrating how they affect both access to AT and the extent to which families’ voices are heard. Figure 1 provides a summary of the power dynamics that we detail in the following sections.

5.1.1 Loss of definitional agency: regulation of AT boundaries. The families we interviewed regularly engage with healthcare providers,

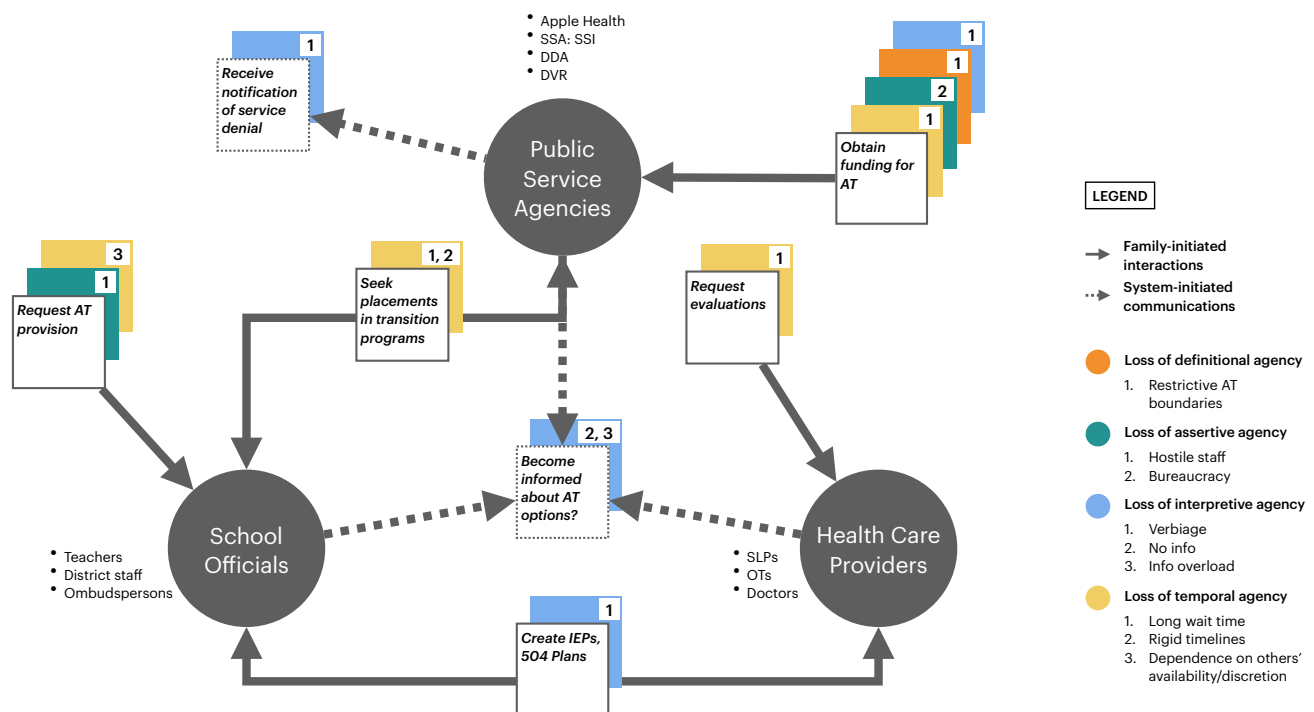


Figure 1: Illustration of how interactions with key institutions (schools, hospitals, and funding agencies) can lead to different forms of agency loss for families, including definitional, assertive, temporal, and interpretive. Solid arrows represent interactions initiated by families to obtain accommodations, while dotted arrows represent information or notifications from institutions that impact access to AT. Colored boxes correspond to the type of agency loss, and the numbers within each box list the specific cause(s) of that loss.

insurance companies, and school districts to obtain funding for assistive devices. In practice, these organizations act as gatekeepers, determining what qualifies as AT through evaluation processes that tend to favor familiar, narrow understandings of disability. This creates uneven access to AT: those with more widely recognized access needs have better access to AT, while those with less conventional needs are left to find their own solutions.

For example, some ATs designed for children are regarded as just “toys” by industry providers, who may lack knowledge or expertise about how different tools may support disabled children. One such AT is the spin disc, a rotating platform a child sits on to spin in place. As noted by the AT specialist and CM4, spinning activates the vestibular system and are particularly useful for neurodivergent children in helping them wind down after a stressful day at school or get ready for bedtime.

Case manager CM4 described a time when she tried to help a child with autism receive financial support for a sensory sit-n-spin seat from a public funding program through ODMF: “I have this one family [...] their child likes to be on the floor all the time. They wanted buy a spin disc, but it’s \$250. The parents couldn’t afford it. We couldn’t buy it in my department either because it looks like a toy. But it’s not [a toy]. It got denied many times.”

The “toy” designation extends to other therapeutic items as well: “Like weighted pillow — it’s meant to give the kids a sense of security: when you put it on your legs, it makes you feel your legs, and it helps you stay sitting for longer periods. However, they will not see it as an item that the child needs. They see it as a toy. The bouncing ball, too. The bouncing ball helps kids who have motor needs [...] I was like, no, there has to be a way that you guys can buy it and give it to the families.” (CM4)

Certain educational items are also perceived as toys even though they play an important role in a non-verbal child’s learning: “I was trying to give one of my families a [talking flashcards] tablet, where you put a card on the back and it will tell you, say, an apple. It’s amazing. And when I tried to get them for my families, they still said no that’s a toy.” (CM4)

Other times, obtaining financial support for AT from funding sources, such as government programs, is complicated by unclear boundaries of AT itself:

“An iPad, they won’t buy it, because it doesn’t come with the applications [already installed] to use it as a communication device. Families have to buy the applications separately, right? And a lot of the applications you have to pay monthly fees. So it’s not something that you can just download to your device and you’re done.

We will not buy a tablet for the families. We will not buy the application, either, [even] if they already have a tablet.” (CM4)

Even though iPads with communication apps such as Proloquo2Go are considered the de facto standard for non-verbal communication, their rigid funding structures leave little room to accommodate differing cost models for hardware and software, further hindering AT access.

These competing criteria for what counts and does not count as AT make AT eligibility an elusive pursuit for low-income families. They are consistently at the mercy of insurance companies and other funding agencies who set the terms of inclusion for AT support, and in doing so, produce normative differentiations between legitimate and illegitimate differences.

5.1.2 Loss of assertive agency: constant flow of judgment. In navigating various services, families frequently face a steady stream of judgment, dismissal, and bureaucratic inertia that create feelings of anxiety, dread, and unease. Over time, the cumulative emotional toll erodes their confidence and willingness to voice or advocate for their needs, leading many to internalize rejection as inevitable.

Interactions with complex bureaucratic systems can be emotionally costly. Parents are often forced to navigate processes that do not acknowledge their expertise in understanding their own or their children’s needs. These systems can silence families by framing their requests as burdensome or troublesome. The violence in this dynamic often manifests in the slow wearing down of families’ resolve and their capacity to advocate for their children. As a result, parents are left feeling powerless, with a fearful orientation to navigating the system, unsure if their voices will ever be truly heard or if they will simply be exhausted into compliance.

When applying for AT support, families can be subjected to experiences that are demoralizing and stressful. For example, they can get asked intrusive questions:

“A lot of people that work at SSI, they don’t care about you or your child, and they don’t care about your feelings. They can be very mean. They ask questions like, ‘What do you need the money for?’, ‘What are you gonna do with this?’, ‘Do you really need to do this and that?’ [...] A lot of parents are like, ‘They’ve made me feel like I’m taking money from their paycheck [...] and they make me feel humiliated.’” (CM4)

At school, parents can also face repeated rejections of accommodation requests for their child — such as a tablet for classroom communication — even though they are legally entitled to these accommodations. CM4 mentions IEP meetings where the psychologist and the special education teacher will say, “I don’t get paid enough for this”, preferring to convince the parents to bring their own tablet than to fill out all the paperwork needed to acquire one for the classroom. This hostility is felt by the parents:

“But a lot of my parents, they will start crying, because ‘if they feel like this, and they say things like that in front of me, what did they do when I’m not there? What did they do with my child? Do they care about my child? Do they treat my kid differently than other kids?’ So

it’s very upsetting, walking into a meeting like that.” (CM4)

Through these experiences, families feel dismissed as nuisances, with their dignity undermined in the process. This fosters a fear-based relationship toward help. Many times, the parents just want to give up:

“A lot of the parents get discouraged. They just want to give up. They want to be done and to try something different. They don’t want to try for what they’ve been waiting for. It has happened with clients where they’re waiting for a communication device from AbleNet. And a lot of families, [worn down by the process], are like ‘I’m done. If it doesn’t happen, good.’” (CM4)

These persistent experiences of rejection and humiliation dissolve families’ willingness to fight over time. Instead of reaching out to inquire about the status of their benefits application, for example, families often just assume they do not qualify and do nothing. After all, it is hard to keep fighting when every effort feels like a losing battle.

5.1.3 Loss of interpretive agency: information opacity and asymmetry. Institutions often operate in opaque and convoluted ways, with little transparency around decisions concerning what families should access or know. This gives institutions disproportionate power over what information is shared and how it is presented, leaving families with little say in what they are allowed to understand or pursue. As a result, families are left navigating a maze of bureaucracy and jargon, unable to make fully informed decisions about AT. This not only undermines their sense of control, but also further entrenches the barriers to using AT effectively.

One such form of opacity has to do with the knowledge required to articulate needs. Obtaining the right AT is not simply about gaining access; it is also about knowing which device is appropriate for particular situations. For many families, the tools needed to make such determinations simply do not exist.

Epistemic gaps can appear on multiple levels. Beyond language barriers, families often lack the knowledge needed to navigate available resources. Many enter meetings with case managers unsure of what to ask for, or even what kinds of support might be available or helpful to them: “To be honest, I didn’t know what to expect. I really didn’t have expectations. I was looking for help because I was feeling kind of lost. And I was desperate. I didn’t know where to start. And that’s how I started. I actually asked, ‘You know, well, what resources do you have? How can you help me?’” (P6).

Although not having interpreters during school or medical visits creates significant barriers for non-English-speaking parents, gaps in understanding can persist even when interpreters are present: “I always ask [the parents], ‘Do you understand what they mean by that?’ Even if you have an interpreter and [they] translate words exactly how the other person said it, that doesn’t mean you understand what they’re saying. Because if you have not been in the system, and you’re new to this, you don’t know a lot of the terms that they use” (CM4).

In short, understanding the system is different from speaking the language. The system has its own vernacular. As CM2 puts it, “language access shouldn’t just be for people that are, you know,

English as a second language. It needs to be accessible for people on all educational levels". Language can become a barrier for both English and non-English speakers, not only in translation, but in how information is communicated:

"The way that verbiage is put in the language of terms and conditions and processes ... can be spoken above someone's head ... above their educational level, above the vocabulary that they've been introduced to, you know, above their glossary." (CM2)

In many cases, families are denied services without understanding why: *"I did it exactly how I understood it, you know?"* (CM2). Often, they only discover afterward that forms need to be filled out differently, in ways that are not specified explicitly on the form. As CM1 pointed out, *"the biggest challenge is awareness [...] The difficulty is really about the inability to ask the right questions"*

As a result of these translation gaps, systems intended to support families remain opaque. Although a wealth of information about AT exists, there are few infrastructures to help families navigate these complex systems. The information they do receive is fragmented and incomplete, at best. Far too often, the families are unaware of ATs that could meet their child's specific needs, such as the eye-tracking communication device from Tobii DynaVox. CM1 wondered: *"Who should be the first person responsible for communicating the existence of the device and its functionality to the family?"* Should it be the AT specialist, the family doctor, or the insurance company? Even when useful information reaches families, it is often delivered incompletely or inconsistently:

"I would say the help that we've received from the school is good but it was limited [...] They said, 'Your daughter needs to find some therapists'. They only gave me the information, but the rest is completely up to me to contact [people] to find the solution." (P11)

In many school districts, a single AT specialist is responsible for multiple schools across the district. Resources are scarce for a single person tasked with managing every student's IEPs and their AT needs. Given these conditions, acquiring the right AT feels like a chance game.

Opacity also manifests in case managers' decisions to simplify the complexities of navigating the system. For example, the AT specialist recounted a time when a DDA case manager assisted a Dari-speaking family in obtaining a communication device. The process was complex, involving multiple stakeholders and separate applications, each with its own timeline and service provider. To avoid overwhelming the family, the case manager intentionally withheld system-level details, such as the need to coordinate across independently operating services. While this approach was intended to reduce stress, the resulting lack of transparency ultimately deepened the family's confusion and sense of powerlessness. After five months of waiting with minimal information or updates, the AT specialist intervened to clarify the process and follow up, without which the family may have abandoned the process altogether.

5.1.4 Loss of temporal agency: (re)production of chrononormativity. Families often have little control over their own timelines, as external forces, such as funding cycles and administrative processes,

follow rigid, predefined clocks. These systems dictate the "when" of AT access: *when* a device is deemed appropriate, *when* funding is approved, *when* referrals are processed, and *when* training is scheduled. Rarely do these institutional rhythms align with the developmental needs, daily routines, or embodied experiences of disabled children and their families.

This loss of temporal agency — the ability to determine the pace at which one learns, adapts, or decides — can lead to a deep sense of powerlessness. One mother (P6) describes calling every organization listed in a folder her doctor had given her, only to be met with months of silence and long wait lists. During those months of waiting, her daughter's condition steadily worsened. Other parents similarly described being trapped in limbo by diagnosis-based access policies, forced to wait for the "right" referral before becoming eligible for devices, even when they already knew what would help their child.

Practitioners, too, are caught in the pressures of externally imposed temporalities that leave little room for care practices. One SLP noted that applying for device funding required unpaid hours of documentation and continuing education, time they simply do not have under current billing codes: *"Getting a device funded requires a lot of written documentation, time that we're not paid for and can't bill insurance for, based on the way our codes are built up [...] It also involves all of the hours I spend understanding the process, being up-to-date with funders and the application process. So most SLPs ... even if they wanted the knowledge, maybe don't have the time, or don't see it as being cost effective."* PM3 echoed the time balance point from the perspective of teachers, describing that they might be hesitant to install a high-tech communication device because of the time it takes to *"implement the system ... and download the right images"*, time that they may not have.

These temporal frictions are deeply consequential. For families, delayed access to AT can stall development during critical periods, especially for children already experiencing delays. P4, described her son with severe autism who missed learning key skills in school. Now at 22, he is too old to return to the regular curriculum and has aged out of after-school programs, leaving him in an educational limbo. Even when families find potentially relevant programs, wait lists can stretch so long that professionals themselves warn not to bother applying. As one mother (P9) put it: *"They let me know that it'll [take] too long. They are honest with me — they don't want to make me wait for long"*.

Each institution — schools, hospitals, insurance companies — has its own chrononormative expectations that dictate how a body ought to operate and progress over time. And bodies that move out of step with these institutional clocks are left behind. In such cases, AT use is inextricably intertwined with the tension between lived time and institutional time, creating temporal mismatches that could discipline bodies into failure.

5.2 Asserting Agency Through Minor Resistance

In the previous section, we conceptualize power as efforts to enforce norms, bodily or otherwise, through standards, emotions, opacity, and timelines. We have described how families encounter rigid

systems that exert power by setting terms of legitimacy, discouraging participation, limiting interpretation, and structuring access. However, where there is power, there is resistance [37]. We have observed families assert their own interests and priorities in response to these moments of friction. Sara Ahmed describes such moments as experiences of “coming up against” [3]. We conceptualize *minor resistance* (1) as indexing strategies of survival may seem trivial from a politically consequential perspective. While small in scale, minor resistance shows families exercising their agency against oppressive structures through everyday acts of defiance. In this section, we highlight the surprising power these “weak” forms of resistance have in building toward an infrastructure of resilience.

5.2.1 Selective compliance. Selective compliance refers to the strategies families devise to make themselves legible to external expectations while still preserving their own ways of doing things. As described in the previous section, institutions’ definitions of AT do not always line up with families’ everyday norms and routines. Rather than taking a directly oppositional stance, families may choose to acquiesce on the surface, adapting just enough to meet external demands. What might appear to be giving in can also be a refusal to assimilate.

For example, P4’s son is 22 years old, non-verbal, and autistic. For a mix of economic and practical reasons, he did not grow up using a tablet to communicate. He finds iPad unfamiliar and does not like to carry it around with him. Over the years, he has primarily used simple sign language and picture cards to communicate with his family. Although he has occasionally used an iPad in school, when he recently was able to take one home for a full year, he still barely touched it. At home, he prefers sign language:

“To be honest, I like the effort he makes, trying to tell me things with his mouth and gestures without using the iPad,” his mom shared.

For him, communicating through the iPad is more cumbersome, since searching through icons to find the right expression takes considerable time. As his mom explained, *“It’s easier for him if people would understand his sign language.”* If they can have things their way, the mom would have preferred a sign language interpreter at school instead of communication tablets. Her son has recently begun using the iPad more regularly over the past six months. Even though learning to use it at this stage is difficult, they believe it will support his communication with teachers and friends at school. At home, however, their communication routines of signs and pictures remain unchanged.

What stands out in this case is that the AT was not meant to “fix” a faulty mode of communication. At home, a better and preferred system already exists; the tablet was used primarily to accommodate others than to alter the family’s core communication practices. This choice can be read as a form of resistance, as it diverts the enforcement of communication norms away from their most personal spaces, such as the home. By engaging only superficially with external expectations, P4 is able to preserve their existing routines in an unobtrusive way.

When it comes to communicating through the iPad, the goal for many of the families is not to increase fluency, as is often assumed. Families often use it as a socially responsive tool, as a way for their child to be legible to others, rather than changing how

communication happens at home. As P1 put it, the goal was not for her son to master the iPad per se, but *“more so that he can use it with other people who don’t understand his language ... or don’t understand his communication”*. This illustrates how AT adoption can be motivated not by a search for better fit but by the need to perform social expectations. Importantly, the family does not fully assimilate the iPad into their daily routines, choosing to use it on a as-needed basis. For the family, what is useful might look like learning just enough vocabulary and not achieving fluency.

Selective compliance is about people intentionally choosing when, where, and how to engage with external norms on their own terms, in ways that prioritize existing linguistic and cultural routines. It is a boundary-setting move that resists the pressures to assimilate.

5.2.2 Strategic refusal. Strategic refusal describes the ways families opt out of available and seemingly helpful resources, not necessarily out of disinterest, but as a negotiation of what is safe, manageable, and worth the risk in their daily lives. Rather than take up every opportunity for support, families frequently assess what forms of help may actually come with hidden costs.

One example of this is the disuse of communication tablets. A common reason for disuse is the fear that their child might damage a borrowed device and that they would have to reimburse the organization that lent it to them. For P9, the reluctance to use the school-issued iPad at home was not due to dissatisfaction with its functionality, but the stress that accompanied its use: *“I think that [the iPad] works very well. If I go purchase that myself, I will be more relaxed, because I will see that if [my child] bites it or something happens, we will not be scared to get charged for that.”* She added that at home, she only lets her 3-year-old son use the iPad when she’s able to supervise him 100% of the time. In this case, interventions that are meant to help can sometimes create more stress than relief. What the families are resisting are not the interventions themselves, but the *terms* in which which they are offered. This specific fear about damaging devices gestures toward a broader theme of how risk gets factored into families’ decisions about whether and how to engage with ATs.

P6 described a similar calculus in deciding not to borrow a weighted vest from ODMF’s lending library, fearing that forgetting to return it could result in additional costs for the family later on: *“Sometimes when you are a single mom that has other responsibilities and everything going on with the other children, it was just one more thing that I did not want to take care of”*. Instead of taking on that risk, she opted to find solutions *“some other way”*. These workarounds may not promise long-term or optimal outcomes (e.g. trying out different potentially useful devices and getting funding for them), but they grant more control in the moment. As she reflected on the emotional toll of these small decisions accumulating over time, she added, *“It made me not want to ask for more help”*. This line shows that for families operating with already stretched resources, help often comes with a price tag (both literally and metaphorically) — guilt if something goes wrong, fear of financial consequences, and exhaustion from navigating complex systems.

This constant tension of juggling competing priorities is particularly salient for P3 whose son has a rare genetic syndrome affecting his mobility. She described having to weigh every AT consideration

Type of Resistance	Trigger / Cause	Power Dynamic Resisted	Outcome	AT Use Pattern	Tactics
Selective compliance	Loss of definitional agency	Pressure to assimilate or conform	Preserving existing routines and practices	Intermittent use: “Using this for you”	Time-boxing; Place-boxing
Strategic refusal	Loss of assertive agency	Participation in systems and processes that are costly or harmful	Asserting autonomy over own priorities	Disuse: “Not using this”	Opting out; Abandoning; Hedging
Covert disobedience	Loss of definitional agency ; Loss of temporal agency	Authoritative claims about what is right or best	Defending lived expertise	Counter-use: “Using that instead”	Borrowing; Second opinion
Folk bridgecraft	Loss of interpretive agency ; Loss of temporal agency	Knowledge silos and temporal discontinuities produced through organizational structures	Building flexibility into rigid systems	Adaptive use: <i>Infrastructuring workarounds</i>	Stopgap solutions; Knowledge brokering; Peer-to-peer support

Table 1: Overview of minor resistance strategies

against basic needs. When given a list of recommended items from an occupational therapist, she recalled: “*They said, okay, these are the things that you can get. But the only income that I have right now is SSI, so to buy everything that he needs on a regular basis is getting so hard for me.*” Sometimes items such as corrective shoes had to be passed over in favor of food and medication: “*I didn’t get the shoes for \$80-\$85 if I have to spend on food [...] I have to manage the money very, very hard.*”

Across these examples, what looks like passive disengagement is in fact a careful and strategic assessment of costs, tradeoffs, and consequences. When margins are thin, families become experts in identifying which forms of “help” are truly helpful, and which may ultimately cost more than their benefits.

At first glance, avoidance as a tactic may seem passive. However, strategic refusal actually involves a clear-eyed identification of the sites where power is exercised. And this awareness is *strategic* precisely because it allows families to navigate power dynamics on their own terms.

5.2.3 Covert disobedience. Covert disobedience describes how families intentionally deviate from official guidance — whether from schools, therapists, or healthcare providers — in favor of lived, experiential knowledge. Compared to previous tactics, this form of disobedience can be seen as the most direct challenge to authority, as it involves a departure from the prescribed course of action and a rejection of the belief that professionals always know what is best. Central to this approach is the belief that institutional expertise does not always align with lived realities. Parents and caregivers may not always feel empowered to challenge authority directly, but they exercise their own judgments in ways that reflect a deep attunement to their child’s needs. These acts of refusal usually take the form of second opinions or parallel paths.

For instance, P10 described how she lost trust in her child’s doctor after being told there was no need for further neurological testing. Not accepting this guidance, she sought out another provider, one

who eventually offered a diagnosis that unlocked access to critical assistive devices at her son’s school.

Similarly, the AT specialist shared a case where her client, a mother of a 9-year-old girl with Down Syndrome, disagreed with the treatment plan of her speech therapist. At 9 years old, the child was still unable to form sentences, confused pronouns, and had difficulty expressing her wants and needs to people outside of her immediate family. She had used an AAC device in the past at school and in speech therapy, and her mother reported seeing a lot of improvement in her verbal communication. However, in the past six months, her mother reported that the speech therapist had stopped using the AAC device, and was “*only working on verbal communication*”. She expressed concerns about discontinuing the use of AAC, because she saw the progress her daughter was making. However, she was reluctant to directly challenge the speech therapist. Even so, she was happy to borrow an AAC device from ODMF to use at home. In doing so, she created a parallel path that honored her own knowledge and observations without direct confrontation.

The impulse to seek alternatives also shows up in how families navigate systemic barriers. One SLP noted that school district-level recommendations are often shaped more by cost and convenience than by individual fit:

“There tends to be, like, one or two types of devices that the school district just picks, usually based on cost. What they recommend for all kids [...] There are very few kids that get something different from that. So typically it’s an iPad with one of two apps, usually Proloquo or TouchChat.”

Recognizing this one-size-fits-all approach, many families pursue outside AAC assessments to access tools that better reflect their child’s needs:

“Even if the school said, ‘this is the device we recommend’, the families are catching on that it comes from a pool of maybe one or two [devices]. So they’re also doing

their own assessments to have a device that's theirs — one they control, can edit, and do whatever they want with [...] They want to have a device that's prescribed more based on medical and clinical necessity than ... a one-size-fits-all approach.” (SLP)

Rather than waiting for school districts to become more flexible, families are proactively finding ways to resist external pressures to normalize.

These workarounds push back against the idea that institutions are always the most reliable, or only, sources of support. They reflect families' ongoing efforts to resist being constrained by rigid, often resource-limited infrastructures. By doing so, families position themselves as experts in their own right. Even though they cannot change the system directly, their deviations reassert the legitimacy of other ways of knowing, understanding, and living with disabilities. Covert disobedience, then, is not simply a rejection of authority, it's an infrastructural critique rendered through practice. It highlights the inflexibility of systems and shows how everyday acts of adaptation and refusal keep families afloat within them.

5.2.4 Folk bridgecraft. Families often find themselves bridging gaps created by systems, developing stopgap solutions that give them ad-hoc access to the ATs they need. These gaps might include slow processes, incomplete information, or bureaucratic hurdles, and rather than working within the confines of these systems, families seek temporary solutions to address their immediate needs.

Take P12, a father who had been waiting for months for an approved device to help with his son's mobility impairment. Frustrated with a system that moved to slow, he adapted an Xbox game controller to build simple games that helped his son exercise his fingers.

Other forms of bridgecraft focus on coordinating or supplementing different forms of AT support. Many parents, faced with gaps in knowledge or experience, end up having to become experts in a variety of topics, from medical information, technology advancements, to insurance policies. P6, for example, described how she needed to become an expert in everything related to her child's care, including learning about how different ATs worked, sometimes learning relevant medical concepts to understand which treatment plan might work best. She also played the role of communicating between speech therapist and the occupational therapist, ensuring that their recommendations did not lead to overlapping AT interventions. With a lack of cross-coordination between these services, she felt that the burden fell on her to stitch together a care plan that would be most beneficial to her child.

This bridging work also includes into collective action. PM3 emphasized that much of the existing AT support infrastructure, especially for those who fall outside the purview of usual channels for support, is parent-built. She shared: *“That's why most nonprofits in the disability field are started by parents. [name redacted], for instance, is a mother whose son has autism. English is her second language, and this whole organization started with parents sitting around a kitchen table, sharing needs, identifying gaps in services.”* PM3 explained that formal institutions, especially schools, tend to offer a narrow set of AT options, usually only for students with very specific needs:

“Because if you have a significant need, or if you're medically fragile or highly impacted with behavior and disability, your pathway is a bit more supported ... and it has a little bit more structure around it. [But] if you're a young adult who doesn't qualify for DDA services (but maybe you're DVR eligible), and you're a bit more cognitively aware and independent, the path is a lot more murky.”

In these cases, families rely on grassroots groups to share knowledge, swap devices, or navigate informal channels to get what is needed. These parent-led networks operate as parallel infrastructures, filling in where institutional support stops short.

By creating their own solutions and pushing the boundaries of available resources, these families are not simply coping with gaps in the system but also actively expanding the possibilities of what can be achieved with what is available. The work these families do in these contexts often go unnoticed and undervalued. While professionals might specialize in one area, families integrate knowledge across multiple domains and networks, creating an infrastructure of care and making that exist outside formal systems, which serves as a lifeline amid systemic shortcomings.

6 Discussion

In the fit model of adoption, technology is framed as the source of empowerment, constitutive of power itself. It casts people as passive recipients of technology and filters diverse AT practices through a binary lens of adoption or abandonment. By contrast, the choice model draws from a productive view of power [37], locating power not in the technology but in how people “name [their] own reality” [19] and negotiate their position within shifting power relations. From this perspective, AT adoption and abandonment are artifacts of sociopolitical processes in which hierarchies of privilege and economic interests are continuously refracted, keeping certain groups at an advantage and others at a disadvantage. Amid this dynamic power terrain, families enact a wide spectrum of practices beyond use and non-use — such as intermittent use, adaptive use, counter use, and disuse — that reveal empowerment as something exercised through action, not simply conferred by technology.

Thus, responding to challenges in AT adoption also requires confronting the harms embedded in the broader infrastructures surrounding use, including fragmented services and pervasive uncertainty. In low-resource settings especially, AT adoption is a “wicked problem” [83], entangled with structural complexities that eschew simple solutions. This calls for developing a critical consciousness around the sociotechnical nature of access work. This is especially important because minoritized communities often have to engage more — not less — with bureaucratic systems like healthcare, welfare, or immigration. As Herd & Moynihan [47] point out, the complexity of government systems (paperwork, eligibility checks, etc.) disproportionately burdens those with fewer resources. Their work, along with our findings, highlight the need for a power-aware perspective in accessibility research and design. Such a perspective asks us to: (1) examine how our own research practices may inadvertently reproduce or sustain power asymmetries, and (2) imagine and build ways for design to transform power relations by amplifying existing practices of minor resistance.

6.1 Questioning Power Relations in Research Practices

In this section, we engage with the normative aspects of accessibility research, interrogating the different ways power operates through the methods and frameworks of our field. These reflections serve as entry points for rethinking whose interests are prioritized and what forms of access are made possible through our research practices.

6.1.1 *AT is not a neutral tool for access provision.*

Q: Who benefits?

The story of P4's son using the iPad only at school and preferring sign language in all other settings is notable because the communication app in his case is not fulfilling a specific need of his — he is not familiar with it and can already communicate effectively without it. Rather, it compensates for the school's failure to provide a sign language interpreter, which calls into question: who is the AT helping, anyway? This case lays bare the power imbalances inherent to the situation. The communication device inscribes power relationships. It constructs P4 as the "other" [100] in relation to the allistic instructor, needing to adopt a mode of communication that the instructor understands. The anecdote challenges the assumption that AT functions solely as an accommodation for disabled individuals, exposing how it sometimes *accommodates* the institution who is unable to provide the appropriate resources.

From this case, we can see that AT is not a neutral tool for access provision — it is an instrument through which disability and communication norms are articulated and enforced. The process of seeking and using AT often comes up against deeper power dynamics, such as who is deemed as deserving of support, who must justify their needs, and who sets the terms of access. Rather than assuming AT straightforwardly empowers its users, we may instead ask "*cui bono?*" [101]: "Who benefits? Whose needs and interests are prioritized?" In this study, AT is the means by which institutions operationalize accommodation. Disability scholar Margaret Price [81] has critiqued individual accommodations for creating a two-tier system. Those with more privilege — who can articulate their needs in institutionally legible ways — are more likely to receive accommodations and succeed within the system; those with less predictable or less socially recognized disabilities are more often excluded. This system is further bolstered by strategies such as "slow-rolling" [79], "rationing by hassle" [34], or "brick walls" [4] to preserve institutional norms and further delimit access. Extending prior research on how AT can perpetuate ableism [12, 96] and exclude disabled voices [100], we argue that AT also mediates institutional logics that normalize the very exclusions it seeks to dissolve.

Instead of understanding AT as a neutral technology, we might consider it as a boundary project [43]: a contested site where competing interests and epistemologies are continuously negotiated. This lens makes clear that every access question must also be a power question. As a complex phenomenon [33] fraught with politics of difference, access is not just about solving a design problem, but also about navigating competing definitions of what counts as legitimate need, desirable behavior, or productive participation.

6.1.2 *Empowerment means being able to define own AT pathways.*

Q: Who has the power to decide AT boundaries, and how is it contested?

Traditional AT adoption narratives conceptualize choice in utilitarian terms: people are expected to select the option that maximizes personal utility [104]. This view posits that the solution to AT abandonment is through discovering, characterizing, and designing for diverse user needs. The logic is that the more diversity we account for, the better options people have to choose from. However, this approach assumes that the *conditions* for meaningful choice already exist. In practice, resource disparities can erase diversity of options downstream (e.g. when schools offer iPads as the sole AT despite the existence of other communication methods). Upstream, traditional user-centered design methods can also give designers unreviewable power to define AT boundaries based on inherited modes of understanding disability, thereby determining which needs matter and deserving of solutions. What is needed, then, is a shift from designing for diverse needs to interrogating the conditions that make diversity possible in the first place [49].

Empowerment is not just about selecting the "right" AT but about being supported in defining one's own AT pathways. This requires asking infrastructural questions [102]: In what ways might our research ecosystem limit the forms of diversity that are possible or thinkable? Where does the sociotechnical system break down in its ability to accommodate the "consequence calculus" [66] that disabled people navigate daily? Can our research methods effectively surface plural practices, including ones not anticipated by designers? How might we train designers to attend to difference?

One way to cultivate designerly attunement to difference is through a "turn to practice" [61], widening our aperture of study from "momentary and historical ... snapshots" of interactions to real-world practices, such as how families design custom AT pathways by stitching together digital and low-tech tools. These practices can serve as starting points for interventions, whether by creating new tools that integrate into existing workflows or by amplifying less visible workflows. Recent work on crip technoscience [42, 50], interstitial practices of care [13], and multilingual accessibility [29] illustrate what this shift could look like. Yet, a practice orientation remains nascent in accessibility research. With this study, we contribute to advancing this turn by foregrounding everyday tactics of resistance as resources for design.

6.1.3 *Recommendations cannot be divorced from structural context.*

Q: Who is doing the work of making AT work?

Very often, interventions around AT adoption and abandonment are proposed without full awareness of the sociotechnical systems in which they are embedded. AT training, for example, is widely cited as a key factor to long-term AT adoption [10, 20]. While training and ongoing support can increase regular use, these interventions are mediated by labor, resources, and time that circumscribe what forms of help are possible and who is expected to provide them.

Consider, for example, the question of who is responsible for delivering AT training — should it fall to the school staff, SLPs, or occupational therapists? In this study, these individuals are frequently overburdened and under-resourced. The SLP we interviewed described the unpaid labor involved in supporting families with AT,

such as keeping up with funding processes, staying up-to-date with ATs, and preparing insurance documentation. Following Star & Strauss's theorization of visible and invisible work [103], we argue that the effectiveness of any intervention depends not only on the quality of its design but also on the labor structures that sustain it. Many solutions are often only viable for those with the economic resources to support them.

AT recommendations must thus be grounded in considerations of the structural conditions in which they will be implemented. Even well-intentioned solutions can inadvertently reproduce existing inequities if the labor they require is invisibilized and under-supported. This raises broader questions for accessibility and HCI researchers: what responsibility do we have for the downstream effects of our interventions? For example, the demands of academic chrononormativity shape the kinds of translational work research projects are allowed to pursue, including how research findings get turned into tangible benefits for the communities engaged in the research. Market forces can privilege some ATs over others: for example, specialized AT such as braille displays remain expensive due to relative low demand, while mass-market accessibility features on Apple devices are comparatively cheaper but tend to flatten the long tail of disability. DIY technologies, intended to bridge the gap between universal low-cost commodity solutions and expensive personalized ones, can still fall short of their democratizing potential because they “require a high level of technical expertise as well as financial investment” [48]. These structural and market forces limit the real-world impact research can reasonably have. There are no easy solutions. Addressing them will involve collectively studying up [69] institutions, rethinking how we structure them, how we evaluate success and failure in research, and which values we commit to uphold.

6.2 Transforming Power Relations Through Design

This study moves beyond tracing how power flows through the AT pipeline to showing how families resist it. Recognizing *power-to* as a core dimension of empowerment for accessibility research invites examination of how computing might be alternatively envisioned to transform power relations beyond mediating them. Minor resistance refers to tactics for negotiating the costs that emerge from contending with power dynamics. This shifts design goals toward creating infrastructures that lower the cost of choice. We explore this by identifying different types of costs families have encountered in this study and offering a few sketches of how community-engaged design might help address them.

6.2.1 Build infrastructure for collective action.

Q: How might we mitigate the cost of compromised choice?

Most choices families make around AT are shaped by structural inequalities and fundamentally compromised [62]. To navigate various constraints, families draw on care networks that consist of friends, advocates, and specialists to share workarounds. These informal forms of infrastructuring [110] sustain families when institutional support is absent. Design can support and amplify these practices by scaffolding local community networks through the creation of online communities that track availability of loaner devices, facilitate repair and adaptation, or enable peer support.

These community-centered approaches point toward the potential for scaling mutual aid into broader social change. Disabled families continue to face significant bureaucratic hurdles in accessing AT, despite international frameworks such as Article 9 of the UN Convention [70] and A/HRC/58/33 [71] mandating accessibility as a human right. In the U.S., where this study took place, state-by-state federally funded protection and advocacy organizations fight for these rights, but they often focus on the most severe cases (such as the overuse of solitary confinement against disabled individuals [84]). Accessibility research can take up the work of supporting grassroots organizing to enable larger systemic change. Research in accessibility-adjacent fields has explored technology's ability to support social movements [40, 60], including connecting people to each other and actions they can take [67], supporting people as they collect data [56–58], documenting problems [30, 31], and studying how communities use social media to organize [60]. However, these projects are not typically designed with accessibility in mind, nor are they implemented in disability-focused contexts. This presents an opportunity for accessibility researchers to explore how research and design can actively support the work of social movements and to consider the roles computing can play in advancing systemic accessibility.

6.2.2 Facilitate collective cataloging of alternatives.

Q: How might we offset the opportunity cost of choice?

Families in this study routinely face difficult trade-offs between immediate and long-term needs. A specialized device might be a good fit technically, but the financial and bureaucratic demands make it inaccessible to many. Opting for a stopgap option, whether a DIY solution or a borrowed device, can also lock families into fragile arrangements. Informed by political theorist and legal scholar Fishkin's notion of *opportunity pluralism* [35], we argue that AT options should be made available across a spectrum of cost and expertise, so that there are multiple “paths to flourishing” — making one choice under constrained circumstances should not foreclose opportunities for meaningful participation.

Such diversity in AT options already exists, largely through community-driven efforts, such as Makers Making Change ² (which creates open-source repositories of AT designs that can be locally adapted and 3D-printed), online forums like r/AssistiveTechnology (which includes a collection of walkthroughs, product mods, and “hacked” AT solutions not found in clinical literature), or social media accounts like @techowl ³ (which shares everyday mobility-related adaptations). However, these sources are scattered and unevenly documented. Also, there are usually no tools to help people weigh different options and make more informed choices. There is a real need for infrastructures and tools that make community knowledge readily discoverable, comparable, and actionable. This might look like building a database of hacks by collating, categorizing, standardizing, and centralizing existing adaptations that are currently strewn across different blogs, forums, or websites, and designing analytic inroads to interrogating and remixing the content. Such a platform can become a collective resource maintained and grown by the community itself.

²<https://www.makersmakingchange.com/s/>

³<https://www.instagram.com/techowlpa>

6.2.3 Devise mechanisms for collective sensemaking of complexity. **Q: How might we reduce the cost of misinformed choice?**

A recurring theme in participants' resistance tactics is the immense labor required to make sense of institutional processes, from special government protocols to insurance paperwork and device procurement systems. These processes produce information asymmetry that undermines interpretive agency: people are being asked to make consequential decisions without adequate tools to assess potential outcomes.

To address this, we call for interfaces that make institutional complexity navigable and investigable. Inspiration can be drawn from projects like the *Usable Privacy Policy Project*⁴ which turns complex web privacy notices into "easy-to-digest format that enables [people] to make more informed privacy decisions as they interact with different websites". This approach could be applied to support informed decision-making around insurance rules, funding processes, and AT information, presenting these materials in ways that facilitate interpretation and evaluation. We might also explore alternative formats, such as annotated guides or interactive comics [109], to make procedural and policy information more engaging and accessible.

Rather than relying solely on automated processes to extract insights from dense texts, interfaces could support collective, participatory sensemaking. For example, MIT's Real Talk for Change project⁵ demonstrates how community members and researchers can work together to interpret complex data, surface shared concerns, and build collective understanding, with support from machine-powered tools rather than driven entirely by them. Similar methods could enable families and advocates to collaboratively make sense of AT documents, preserve interpretations, and structure them so future groups can explore, extend, and adapt this knowledge. This approach amplifies the interpretive labor already being performed informally in communities, centering co-production of insights rather than automated summarization.

7 Conclusion

This paper has examined how power circulates in the everyday lifeworlds of disabled families navigating AT services. Through ethnographic engagement with families and service providers, we highlight how decisions around AT use are shaped by institutional timelines, bureaucratic hurdles, and information opacity. In response, families engage in minor forms of resistance — selective compliance, strategic refusal, covert disobedience, folk bridgecraft — to make space for choices that align with their lived realities.

In this paper, we argue for an approach to accessibility that is attuned to the politics of infrastructure, the labor of care, and the friction-filled conditions under which access must be negotiated. We offer the notion of *power-to* as a way to expand computing's role in empowerment — not only to enable independence and interdependence, but also to support the capacity to define one's own path through harmful systems. This reframing opens up design practices that support communities in building grassroots power through self-determined modes of access.

⁴<https://usableprivacy.org/>

⁵<https://realtalkforchange.org/>

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